

Design and analysis of a long-range business jet

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Executive summary

This report details the conceptual design process for a long-range business jet with small airfield capabilities. An iterative design process was used. Initial value estimates were obtained through a historical study. Sizing from a conceptual sketch was undertaken before completing a more detailed aerodynamic analysis using empirical drag data. The final design iteration comes to an aircraft with a max take-off weight of 84,260 kg, thrust-to-weight ratio of 0.36 and range of 6000nm. The aircraft can take-off and land from a minimum airfield length of 1,400m, allowing it to operate out of 90% of the world's airports.

1. Project brief

1.1. Problem Being Solved

The project aims to address the increasing congestion at major international airports, which are operating beyond capacity during peak periods. As global air travel demand is expected to increase in the next 15-20 years, traditional high-capacity aircraft, such as the Airbus A380 and Boeing 747, are being phased out due to high operational costs and infrastructure limitations. Additionally, smaller business jets, which could alleviate congestion by operating from satellite airports, are often financially inaccessible for most business travellers.

Previous attempts to design business-class-only aircraft have failed to offer a competitive advantage over existing short-range flights. Therefore, the problem revolves around creating a viable business jet solution that can offer international business-exclusive services from smaller airports, easing congestion while meeting the needs of business travellers.

1.2. Product

The product is a new business jet designed to transport 80 business-class passengers and their baggage over a range of 6000 nautical miles at a competitive speed. The aircraft will provide a service level equivalent to or better than current offerings for business travellers and will be able to operate from regional airports. The design will emphasise creating a unique and competitive service compared to existing operations, focusing on the needs of higher-paying passengers.

2. Project requirements

The requirements of the project were determined from [1] and are as follows:

- Range of 6000 *nmi*.
- PAX of 80.
- Climb rate of 300 *fpm* from 35000 *ft* altitude.
- Two pilot and four cabin attendant crew.
- Cruise speed comparable or greater than existing services.
- Operate from regional airports.
- Offer a competitive and unique service.
- Investigate potential future development of the aircraft.

3. Feasibility study

3.1. Payload/range

The required PAX of 80 puts this design on the lower end of commercial airliner capacity. This number is seen more on the likes of regional airlines and turboprop transports rather than long range jets. However, this jet is to operate in a unique space, flying long-range missions while operating out of smaller regional airports rather than larger hub airports.

Long range routes are often limited to the number of airports which can accommodate the large aircraft which fly them. Further, they are limited to airports in cities with larger markets which can meet the demand for larger passenger numbers. To fly long range routes, passengers who do not live in these hub cities will often have to fly to them first, increasing the cost and inconvenience of their trip.

For business class passengers, minimising travel time and inconvenience is a priority. It is likely they will take a direct flight option if it is available. By offering long range travel routes operating out of smaller regional airports, these direct flights would become the first option for business class passengers. Of course, the demand will be lower for these flights in smaller cities hence the choice to adhere to the minimum PAX requirement to ensure the flights can be operating at maximum capacity.

3.2. Cruise speed

Business passengers desire reduced flight times on long range flights. Cruise speed has a significant impact on the total flight duration, especially for long haul routes where the cruise segment makes up an increasingly large fraction of the total mission. Current long haul commercial jets operate at around Mach 0.85. Aerodynamics, especially for larger jets, become increasingly complex above this speed due to parasite-drag build up [2]. The target cruise speed for this aircraft was selected to be Mach 0.85 to maximise the use of current established aerodynamic technology around this speed.

3.3. Cruise altitude

Current long range jets cruise in the range of 31,000 – 38,000 feet [3]. The target cruise altitude for this jet will be set at 35,000 feet. This is a common cruise altitude and thus there is an abundance of property data and technology research on cruise at this height, which made the initial analysis simpler.

3.4. Field length requirements

This aircraft aims to be the most versatile aircraft of its kind in terms of its ability to land at almost any regional airport worldwide. The target percentage of airfield capability is 90%, giving unparalleled access to any almost any market imaginable. The minimum airfield length required to be able to take off and land with this jet is 1,400 metres. However, as field lengths decrease in size usually so too does the airfield infrastructure. A jet with a large wingspan may not be able to be accommodated at smaller airfields due to space restrictions or lack of suitable facilities. More research needs to be conducted into identifying market airfields and assessing their capability of accommodating this jet type to ensure route versatility.

3.5. Alternative roles and development potential

The jets unique capability to land at 90% of global airfields could be utilised strategically in many other capacities and roles alternative to civilian transport. Direct, long-range travel between smaller airfields is a desirable trait of a private jet. Customers of private jets looking to travel similar routes can enjoy a much larger cabin volume when compared to current market options. The cabin can be reconfigured to suit the needs of the customer without trading performance. The jet presents the perfect option for sports team charter. Teams who currently must hire larger jets with less range could have a jet which fits their team size and route needs. The field length capabilities would also be attractive to military operations. Flying at range into small or temporary airfields in strategic locations could be useful for transport of military personal or equipment, or the long loiter time could be used for surveillance and aerial command.

4. Initial estimates

4.1. Empty mass fraction, fuel fraction and maximum take-off weight

A statistical analysis of ten different aircraft with a similar mission profile (simple cruise) to the project was undertaken is presented in Table 4.1. Aircraft were selected primarily based on their similarity to the required 6000 *nmi* range.

Table 4.1: The performance of competitor aircraft.

Aircraft name	PAX [people]	Range [nmi]	W_0 [kg]	W_e/W_0 [–]	W_f/W_0 [–]	T/W [–]	W/S [kg/m ²]	Source
Airbus A330-300	300	6345	242000	0.51	0.45	0.26	669	[4]
Boeing 767-300ER	290	6310	186880	0.48	0.48	0.29	660	[5]
Boeing 767-400ER	409	5625	204116	0.51	0.45	0.27	702	[5]
Boeing 787-10	440	6,430	254,011	0.53	0.40	0.27	673.77	[6]
Bombardier Global 7500	19	7700	52100	0.54	0.41	0.33	447.21	[7]
Boeing 777-200ER	313	5,825	263,080	0.54	0.52	0.32	614.96	[8]
Bombardier Global 6500	17	6,600	45,132	0.51	0.45	0.30	76.08	[9]
Bombardier Global 5500	16	5,900	41,957	0.55	0.42	0.33	442.58	[10]
Airbus A350-1000	480	7,950	311,000	0.50	0.42	0.28	669.83	[11]
Airbus A321XLR	220	4,700	101,015	0.50	0.39	0.30	825.29	[12]
Averages	250	6339	170129	0.52	0.42	0.30	618	

Based on the statistical data in Table 4.1, an initial estimate of max take-off weight was decided to be 65,000 kg. This was done by dividing the average max take-off weight by the average PAX to get an estimate for the average tons per passenger (TPP) as seen in Equation 1.

$$TPP = \frac{W_{0AVG}}{PAX_{AVG}} \approx 55,000 \text{ kg} \quad (1)$$

Ten tons was added to this weight to give a bit a leeway, yielding the initial max take-off weight estimate of approximately 65,000 kg.

4.2. Thrust-to-weight ratio and engine sizing

The thrust-to-weight ratio (T/W) was initially estimated statistically. The line of best fit on a graph comparing the Mach number to T/W of comparable aircraft has an equation in the form of Equation 2.

$$T/W_0 = aM^C = 0.25 \quad (2)$$

Where a is 0.267, C is 0.363 and M is *Mach* 0.85.

The $T/W_{take-off}$ was also calculated using Equation 3 from [2].

$$T/W_{take-off} = (T/W_{cruise})(W_{cruise}/W_{take-off})(T_{take-off}/T_{cruise}) = 0.276 \quad (3)$$

The lift to drag ratio was initially estimated using Equation 4.

$$\frac{L}{D} = 15.5\sqrt{A_{wet}} = 20.01 \quad (4)$$

Where the wetted aspect ratio (A_{wet}) was estimated using Equation 5 from [2].

$$A_{wet} = \frac{A}{S_{wet}/S_{ref}} = 1.67 \quad (5)$$

The wetted area ratio (S_{wet}/S_{ref}) was chosen to be 6 from Figure 3.6 in [2].

T/W_{cruise} is the inverse of L/D_{cruise} . The larger value of 0.276 was multiplied by $W_0 = 65,000 \text{ kg}$ to calculate an initial thrust (T) requirement of 89,561 N needed per engine. Therefore, PW1000G engines were selected that have a T of 92,790 N and a SFC in cruise of 0.52/hr determined from [1]. In sections 7 and 8 of this report, updated values of $W_0 = 87,280 \text{ kg}$ and $T/W = 0.292$ were determined to satisfy take-off distance requirements. Using the updated values, the above calculations were iterated to find a T requirement of 125,195 N per engine. Hence the engine selection was updated to the LEAP-1A which has a T of 143,050 N and a SFC in cruise of 0.51 /hr.

4.3. Wing loading and gross wing area

Initially, a value for wing loading (W/S) of 618 kgm^{-2} was selected from Table 4.1. This could be used to calculate the wing reference area (S_{ref}) using Equation 6 from [2].

$$S_{ref} = \frac{W_0}{W/S} = 107 \text{ m}^2 \quad (6)$$

In sections 7 and 8 of this report the W_0 and W/S were re-calculated as a values of 87,280 kg and 301 kgm^{-2} respectively, to facilitate mass fraction and landing length requirements. An updated value for S_{ref} was then calculated to be 290 m^2 .

5. Geometry

5.1. Configuration

Figure 5.1 illustrates the initial configuration layout of the aircraft showing the main features and a scale bar.

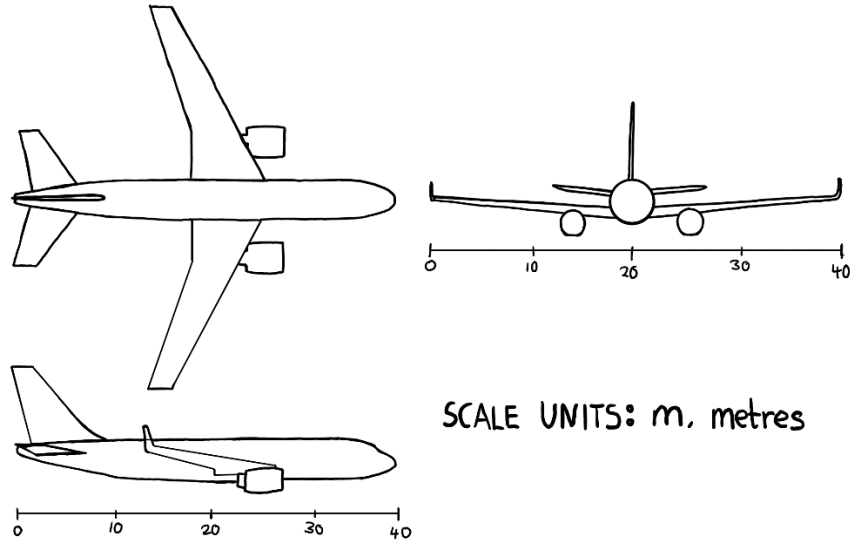


Figure 5.1: Initial aircraft layout.

A wing and rear tail configuration were chosen for this design to ensure aerodynamic stability. The rear tail locates the centre of pressure (the point through which the sum of aerodynamic forces is considered to act) aft of the centre of gravity, which provides a restoring force couple after any variation in pitch as outlined in [14]. Aft-swept wings were chosen because of their proven stability and reliability in civilian airliner history. A cylindrical fuselage with conical nose and upswept tail were also selected due to proven effectiveness and simplicity of approximating drag from historical data.

A low wing was preferred over a high wing due to weight, aerodynamic, and maintenance considerations. High wings must be stronger, thus weigh more than low wing designs. High-wing designs must be accompanied with a T-tail configuration to ensure the horizontal stabiliser is passing through clean air. T-tails have to be stronger than conventional tails thus adding to weight. According to [15], low wings have the capacity to be more aerodynamically efficient than high wings. Lower wings are also easier to maintain as they are closer to the ground for technicians to access. Fuel ports on the wings are also easier to access.

Under-wing mounted engines were selected as opposed to aft mounted engines. The primary reason for this was maintenance considerations. It is much easier to inspect and maintain an engine which is mounted under the wings when compared to an engine mounted higher on the aft of the jet. Special equipment is needed to conduct aft mounted engine maintenance so that technicians can reach the engines. This equipment would likely not be available at smaller regional airports that this aircraft is targeting. Therefore, it makes more sense to have engines which can be more easily inspected and maintained at all airfields it may fly to. Avoiding an aft mounted configuration also means a T-tail does not have to be used which saves weight.

5.2. Wing geometry

The wingspan (b) was initially calculated using Equation 7 from [2].

$$b = \sqrt{AS_{ref}} = 32.74 \text{ m} \quad (7)$$

Where the aspect ratio (A) was chosen to be the upper bound of typical commercial planes with a value of 10 as described in [2]. The aerofoil thickness ratio (t/c) and leading-edge sweep (Λ_{LE}) were determined to be 0.14 and 30° respectively using our cruise speed and figures from [2]. The taper ratio (λ) was decided to be 0.3 from [16]. The quarter chord sweep ($\Lambda_{C/4}$) was calculated using Equation 8 from [2].

$$\Lambda_{C/4} = \tan^{-1} \left[\tan \Lambda_{LE} - \left(\frac{1 - \lambda}{A(1 + \lambda)} \right) \right] = 27.6^\circ \quad (8)$$

The root chord (C_{root}) was calculated using Equation 9 from [2].

$$C_{root} = \frac{2S_{ref}}{b(1 + \lambda)} = 5.04 \text{ m} \quad (9)$$

The tip chord (C_{tip}) was calculated using Equation 10 from [2].

$$C_{tip} = \lambda C_{root} = 1.51 \text{ m} \quad (10)$$

The mean aerodynamic chord (MAC or $\bar{C}$) was calculated using Equation 11 from [2].

$$\bar{C} = \frac{2}{3} \left(\frac{C_{root}(1 + \lambda + \lambda^2)}{1 + \lambda} \right) = 3.59 \text{ m} \quad (11)$$

The perpendicular distance from the centreline to the MAC ($\bar{Y}$) was calculated using Equation 12 from [2].

$$\bar{Y} = \frac{b}{6} \left(\frac{1 + 2\lambda}{1 + \lambda} \right) = 6.72 \text{ m} \quad (12)$$

Once the wings were sized, the fuel tank geometry was determined to find the fuel capacity. A graphical method of interpreting ratios of a given fuel tank dimensions to an existing wing from [1] was used, and these ratios were applied to the design wing geometry. The wing fuel tanks were broken into two sections divided by three cross-sectional areas. The cross-sectional areas (A) are a function of the average thickness of the wing (T) and the width of the tank (W) (which is a ratio of the local chord). Local wing thickness values were calculated based on historical data provided in [1]. The tanks profile volume was calculated as a function of these cross-sectional areas and the perpendicular length of the tank to the centreline (L).

$$A = TW \quad (13)$$

$$V_{total} = 2 \times (L/6)(A_1 + A_2 + 4A_{mid}) = 56.4 \text{ m}^3 \quad (14)$$

The weight of fuel required ($W_{f_{required}}$) was calculated by multiplying $W_{f_{required}}/W_0$ by the initial W_0 , yielding a value of 21,764 kg. This is more fuel than can fit in the fuel tanks. However, in sections 7 and 8 of this report, the W_0 , W/S and therefore S_{ref} were re-calculated to meet mass fraction, take-off, landing and stall speed requirements. Reiterating the calculations with these revised values produced a $W_{f_{required}}$ of 25,149 kg and a fuel tank capacity of 61,484 kg; more than enough. The final wing geometry values can be seen in Table 5.1.

5.3. Fuselage geometry

The fuselage geometry calculations were done for three different seating configurations, (2-2), (3-3), and (2-3). However, only values for the (3-2) configuration are shown in this report.

The seat pitch, seat width, aisle width, galley volume and number of toilets were chosen as 0.97 m, 0.57 m and 0.61 m, 12 m³ and 4 respectively using the upper bound of typical business class passenger compartment data found in [1]. Adding together the seat and aisle widths gave an internal diameter of 3.46 m. An estimated wall thickness of 0.1 m yielded an external diameter of 3.66 m.

The passenger cabin length was calculated by multiplying the number of rows (16) by the seat pitch for a value of 15.52 m. The length need for toilets was determined to be 4 m using a value of 1 m per toilet from [2]. The galley length was calculated by dividing the galley volume by the internal cross-sectional area of the cabin (not including the aisle) to retrieve a value of 2.74 m. The nose fineness ratio could be determined using figures retrieved from [1].

Multiplying the nose fineness ratio by the external fuselage diameter gives a nose length of 6.22 m. The tail angle and length were determined using the average tail angle and length to diameter ratio for transport jets from [1] and multiplying the length to diameter ratio by the external fuselage diameter for a value of 12.08 m. Summing all the lengths together found an overall fuselage length of 40.56 m.

A 3-2 configuration was chosen to enable adequate head room, aisle height and overhead luggage space while maintaining a simple circular cross section. Additionally, the length of the (3-3) and (3-2) configurations are very similar (40.67 m and 40.56 m respectively) because the tail length decreases with decreases in diameter and the 3-2 configuration divides evenly into the 80 total seats required. This results in a similar length without an unnecessary increase in diameter.

5.4. Tail geometry

The aircraft's tail provides stability in the form of pitch and yaw control. Traditional tail geometries are broken into two components: the vertical and horizontal tails. A conventional tail was chosen for this initial design because as mentioned in [2] it provides adequate stability and control authority without sacrificing significant weight.

Traditional design convention discussed in [2], dictates that the horizontal tail's leading-edge sweep should be 5° greater than the leading-edge sweep of the main wing. This is to delay stall on the horizontal tail after a main wing stall, so the aircraft can maintain pitch authority and make a recovery. It is also mentioned that the vertical tail leading edge is usually set between 35° and 55°. The tail thickness ratio t/c was approximated to be similar to that of the main wing.

After deciding the leading edges of the tail, a graphical method was used to determine the shape of the tail sections as a function of their root chord length. Changing the root chord length changes the reference area of the wing sections, which is used to determine the moment arm between the aerodynamic centres of the wing and tail. The moment arms of the vertical and horizontal tail were determined analytically.

$$L_{HT} = \frac{C_{HT} \bar{c} S_{ref}}{S_{HT}} = 21.85 \text{ m} \quad (15)$$

$$L_{VT} = \frac{C_{VT} b S_{ref}}{S_{VT}} = 21.85 \text{ m} \quad (16)$$

Where L_{HT} and L_{VT} are the horizontal and vertical tail moment arms, C_{HT} and C_{VT} are the tail volume coefficients (given as 1.0 and 0.09 respectively in [1]), $\bar{c}$ is the mean aerodynamic chord, b is the wingspan and S_c is the reference area of component c . The reference areas of the tail sections were adjusted until the moment arms fell within 50 – 55% of the fuselage length according to convention stated in [1] and were 56.5 m^2 for the horizontal and 46.4 m^2 for the vertical tail.

5.5. Summary of aircraft geometry

Table 5.1 displays a summary of the final design iteration geometry sizing.

Table 5.5: Final summary of aircraft geometry.

Fuselage length [m]	40.6		
Passenger cabin length [m]	22.3		
Fuselage external diameter [m]	3.7		
	Wing	Horizontal tail	Vertical tail
Area, S [m^2]	230.0	56.5	46.4
Aspect ratio, A [–]	10.0	5.0	1.5
Taper ratio, λ [–]	0.3	0.3	0.3
Span, b [m]	47.9	16.7	8.4
Mean chord, $\bar{C}$ [m]	5.3	3.7	6.1
Leading edge sweep, A_{LE} [°]	30.0	35.0	40.0
Quarter chord sweep, $A_{C/4}$ [°]	27.6	30.6	33.5

6. Aerodynamics in cruise

6.1. Component build-up method

The aerodynamic forces of the aircraft design were determined from the initial sizing, to verify whether further iteration was needed to meet project requirements. To save time and complexity on this initial design phase, only the aerodynamics in the cruise mission segment were considered for analysis.

To estimate the total parasitic drag of the proposed design, the component build-up method was employed. This method estimates the subsonic parasitic drag of each aircraft component and sums them to find an overall aircraft estimation. Parasitic drag due to any miscellaneous geometries as well as any leakages or protuberances was also considered in this method. Equation 17 from [2] shows the summation of these drag components.

$$C_{D_0} = \frac{\sum (C_{f_c} FF_c Q_c S_{wet_c})}{S_{ref}} + C_{D_{misc}} + C_{D_{L\&P}} \quad (17)$$

Each geometric component of the aircraft contributes to the overall parasitic drag. These components have a coefficient of skin friction drag (C_f), form factor (FF_c) to compensate for pressure drag due to flow separation, component interference (Q_c) to account for the mutual interference between components, and wetted area (S_{wet_c}). The sum of these component drags was divided over the reference area of the wing. $C_{D_{misc}}$ and $C_{D_{L\&P}}$ represent the miscellaneous drag and leakage and protuberance drag components respectively.

C_{f_c} denotes the flat plate skin coefficient of a specified component, which is a weighted average of the both the turbulent and laminar flat plate skin coefficients for that surface. Due to the no slip condition and viscous nature of air molecules, air moving at speed across a flat surface will experience friction between is shear layers tangential to the surface. This ‘skin friction’ force will vary in magnitude depending on the nature of the flow. At the leading edge of a surface, the flow is likely to be laminar which results in a lower skin friction drag. As the flow matures across the surface it is traversing, it will transition into turbulent flow, resulting in a thicker boundary layer and thus higher skin friction. The skin friction coefficient is a function of Reynolds number. Analytically, C_f was determined as a function of Re , as well as Mach number for turbulent flow.

$$[Turbulent] \quad C_f = \frac{0.455}{(\log Re)^{2.58} (1 + 0.144 M^2)^{0.65}} \quad (18)$$

$$[Laminar] \quad C_f = \frac{1.328}{\sqrt{Re}} \quad (19)$$

The Reynolds number is a ratio of the inertial and viscous forces within a fluid and was calculated for each aircraft component using Equation 20 from [2].

$$Re = \frac{\rho V L}{\mu} \quad (20)$$

Where ρ , μ and V are the air's density, viscosity and speed respectively. The air density and viscosity are the values for air at a cruise height of 35,000 ft. L represents the characteristic length of the component's surface, which is the length tangential to the flow velocity. Table 6.1 summarises the values for each component's characteristic length and corresponding Reynolds number.

Well-designed aerodynamic surfaces are engineered to maximise the percentage of the characteristic length, which is governed by laminar flow, thus reducing the skin friction drag along such surface. Research done with modern technology suggests that a 50% laminar flow fraction can be achieved for the fuselage and up to 80% for the wings and tails. To be conservative, the established laminar flow fractions of classic production metal for civil jets was used; 5% laminar flow over the fuselage and nacelles, and 10% over the aerodynamic surfaces.

6.1.1. Wing and tail surfaces parasitic drag

To calculate the skin friction coefficient (C_f) for the wing and tail surfaces, the characteristic length (L) of these surfaces were taken to be equal to the mean aerodynamic chord ($\bar{c}$) of each surface. Aerodynamic forces were considered to act across $\bar{c}$ when dealing with length scale, and through the quarter-chord ($c_{1/4}$) when considering forces about a singular point. The weighted average to find C_f was 10% of the $C_{f_{laminar}}$ and 90% of $C_{f_{turbulent}}$.

Form factors specific to aerodynamic surfaces needed to be calculated for each component. FF_c for wings and tails were calculated with the following equation:

$$FF = \left[1 + \frac{0.6}{(x/c)_m} \left(\frac{t}{c} \right) + 100 \left(\frac{t}{c} \right)^4 \right] \cdot [1.34 M^{0.18} (\cos \Lambda_m)^{0.28}] \quad (21)$$

Where $(x/c)_m$ is the chordwise location of the aerofoil's maximum thickness (estimated to be 0.5 for a high-speed aerofoil), (t/c) is the thickness ratio of the aerofoil, and Λ_m is the sweep of the maximum thickness line (approximated as the quarter-chord sweep for each surface).

The components of interference drag (Q_c) were approximated using estimations from [1] to be 1.0 for the wings (well filleted), 1.03 for the vertical tail, and 1.08 for the vertical tail.

The wetted area (S_{wet}) was calculated by doubling the reference area (S_{ref}) for both the horizontal and vertical tail. For the main wing, a graphical method was employed to determine the exposed wing area ($S_{exposed}$), and this was doubled to find S_{wet} .

6.1.2. Fuselage parasitic drag

To calculate the Reynolds number for the fuselage, the characteristic length was taken as the total length of the fuselage. The weighted average to find $C_{f_{fuselage}}$ was 5% of $C_{f_{laminar}}$ and 95% of $C_{f_{turbulent}}$. The Equation 22 below was used to calculate $FF_{fuselage}$.

$$FF = 0.9 + \frac{5}{f^{1.5}} + \frac{f}{400} \quad (22)$$

Where f is the fineness ratio of the fuselage, a ratio of its total length to its external diameter. The fuselage's Q_c was approximated to be 1.0 as shown in [1], and to calculate S_{wet} , the fuselage was approximated as a cylinder of equal length as seen in Equation 23.

$$S_{wet} = \pi D L + \frac{1}{2} \pi D^2 \quad (23)$$

6.1.3. Nacelle parasitic drag

The length of the nacelle was used as its characteristic length in the calculation of Re . There was no known specification of the laminar flow fraction for a nacelle, so it was approximated to be same as that of a fuselage, as they share similar cylindrical geometry [16]. Thus 5% of the $C_{f_{laminar}}$ and 95% $C_{f_{turbulent}}$ was used to find the overall $C_{f_{nacelle}}$. The FF for the nacelles were calculated using Equation 24.

$$FF = 1 + \frac{0.35}{f} \quad (24)$$

Where f is the fineness ratio of the nacelle. The Q_c of the nacelles was chosen using [1] to be 1.5, as they were to be directly mounted in order to minimise the necessary landing gear height. The S_{wet} of the nacelles were calculated using the same cylindrical approximation as the fuselage. Where L was the nacelle length and D was the nacelle external diameter.

6.1.4. Miscellaneous parasitic drags

An empirical method was used to estimate the contribution of miscellaneous objects sticking out into the flow. The $C_{D_{misc}}$ contribution for each miscellaneous component was calculated from using Equation 25 from [2].

$$C_{D_{misc}} = \frac{C_{D_{\pi}} \cdot (Frontal Area)}{S_{ref}} \quad (25)$$

Where $C_{D_{\pi}}$ is a ratio of an experimental determined drag-to-dynamic-pressure value and frontal area of a specified component as seen in Equation 26 from [2].

$$C_{D_{\pi}} = \frac{D/q}{(Frontal Area)} \quad (26)$$

The $C_{D_{\pi}}$ value for a smoothly faired windshield was approximated to be 0.07 from [2]. No other miscellaneous components were considered due to either their absence from the design or the fact that they are ‘hidden’ during cruise.

Another component of miscellaneous drag to consider was the drag due to the additional flow separation caused by the upsweep of the aft fuselage. The empirical D/q relationship for this additional drag was approximated as Equation 27 from [2], which is a function of the upsweep angle (u) in radians, and maximum frontal area of the fuselage (A_{max}).

$$D/q = 3.83u^{2.5}A_{max} \quad (27)$$

D/q was divided by S_{ref} to find the $C_{D_{misc}}$ contribution of this additional drag. The total miscellaneous drag was calculated as the sum of the miscellaneous drag components.

6.1.5. Leakage and protuberance parasitic drag

The contribution of leakages and protuberances to the drag of the aircraft is not easily predicted. Leakages either ‘inhale’ or ‘exhale’ air, directly increasing the drag or flow separation respectively. Protuberances include any odd discontinuities in the aircraft geometry. For civilian airliners these can include antennas, lights, actuator fairings, rivets, door edges etc. As the quantity and details of these specific protuberances cannot be known until much later in the design process, this drag will be estimated as a percentage of the total parasite drag. The range of L&P drag percentage was given to be 2-5% for jet transports as discussed in [2]. The mean of this range, 3.5%, was chosen to be used for the analysis.

6.1.6. Summary of parasitic drags

Table 6.1 shows the final iteration of contributing values for the component parasitic drag estimation.

Table 6.1: Component parasitic drag estimations.

Component	L [m]	Re (E^7) [-]	Cf (laminar) (E^{-4}) [-]	Cf (turbulent) (E^{-3}) [-]	Laminar flow fraction [-]	Cf (E^{-3}) [-]	FF [-]	Q [-]	Swet [m2]	Drag component (E^{-3}) [-]
Wing	5.30	4.09	2.08	2.27	0.1	2.06	1.52	1.00	424.5	3.1
Horiz. Controls	3.74	2.88	2.47	2.39	0.1	2.18	1.52	1.08	56.54	3.6
Vert. controls	6.06	4.68	1.94	2.22	0.1	2.02	1.52	1.03	46.4	3.2
Fuselage	40.56	31.3	0.751	1.71	0.05	1.63	1.06	1.00	487.4	1.7
Nacelles (2 off)	3.33	2.57	2.62	2.43	0.05	2.32	1.23	1.50	53.7	4.3
Misc.										3.7
L&P										0.5
C_{D_0}										17.7

6.2. Induced drag

A large portion of aerodynamic drag is a result of the induced drag caused by wingtip vortices. Wingtip vortices are created from the high-pressure air spilling over the wing tips from below the wing into the low-pressure region above the wing. The rotational energy caused by this phenomenon is wasted energy and contributes directly to drag. The induced drag can be calculated using equation 28 from [2].

$$C_{D_{induced}} = KC_L^2 = 1.81E^{-4} \quad (28)$$

Where C_L is the coefficient of lift at cruise and K is the ‘drag-due-to-lift factor’. K can be approximated using the Oswald span efficiency method found in [2].

$$K = \frac{1}{\pi Ae} = 0.0398 \quad (29)$$

Where e is the non-elliptical lift distribution which accounts for the additional flow separation of non-elliptical wings. e was approximated from [1] to be 0.8.

As discussed in [2], the lift coefficient at cruise (C_L) was approximated by using the aerodynamic coefficient definitions for lift as seen in Equation 30.

$$L = qS_{ref}C_L \quad (30)$$

Where dynamic pressure (q) is defined as

$$q = \frac{1}{2}\rho V^2 \quad (31)$$

L is the lift force of the aircraft during cruise. At cruise, the lift force is exactly equal to the weight force of aircraft. The maximum weight of the aircraft at cruise (W_2) was used to determine the lift coefficient. Rearranged to find C_L , the Equation 32 becomes:

$$C_{L_{cruise}} = \frac{W_2}{qS_{ref}} \quad (32)$$

6.3. Total drag

For an uncambered wing, which for simplicity was used in this conceptual design analysis, the total drag coefficient is the summation of C_{D_0} and $C_{D_{induced}}$ as seen in Equation 33.

$$C_D = C_{D_0} + KC_L^2 = 0.01952 \quad (33)$$

According to Equation 38 for drag force found in [2]:

$$D = qS_{ref}C_D \quad (34)$$

The maximum cruise drag force was calculated using Equation 35.

$$D_{cruise_{max}} = 72,298N \quad (35)$$

Due to the force balance at cruise, the maximum cruise thrust $T_{cruise_{max}}$ is equivalent to $D_{cruise_{max}}$. The ratio of $T_{cruise_{max}} / T_{max_{take-off}}$ using the thrust from final chosen engines (CFM LEAP-1A) is 25%, which matches predicted industry standard values.

6.4. Lift-to-drag ratio

The maximum lift to drag ratio was found at the intersection of the origin tangent to the drag polar. The drag polar is a parabolic curve on the C_D - C_L axis defined by the uncambered drag coefficient Equation 28. Figure 6.4 presents this curve, its tangent, and their point of intersection denoting the maximum lift-to-drag coefficient.

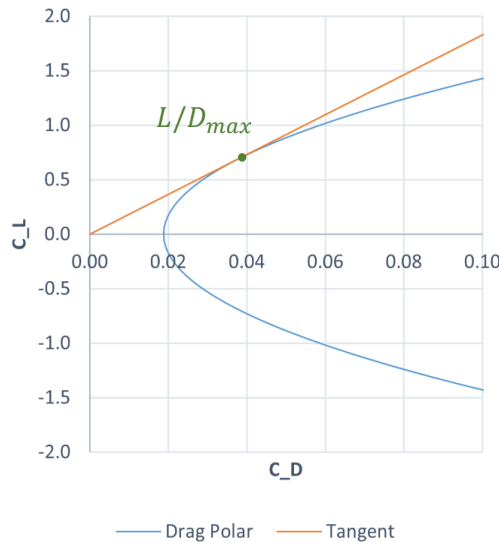


Figure 6.4: Drag polar - tangent intersection to determine L/D_{max} .

The gradient of the tangent line yielded the value of L/D_{max} during cruise, which was determined to be 18.3 which seems reasonable based on the lift to drag ratio of comparable aircraft such as an A320.

7. Masses

Mass fractions were calculated for each component of the aircraft using typical estimate equations. The equations for each component were all found in [1].

7.1. Wing structure

$$W_{wing} = 0.1W_0 = 8,623 \text{ kg} \quad (36)$$

7.2. Tail structure

$$W_{tail} = 0.02W_0 = 1,325 \text{ kg} \quad (37)$$

7.3. Fuselage structure

$$W_{fuselage} = (0.039)(1.08)(0.96)(2L_{fuselage}D_{fuselage}V_D^{0.5})^{1.5} = 15,236 \text{ kg} \quad (38)$$

Where dive velocity (V_D) is 1.4 times the cruise velocity as is commonly used in [1].

7.4. Nacelle structures

$$W_{nacelle} = 0.0068T = 1,262 \text{ kg} \quad (39)$$

Where T is the combined thrust of both engines.

7.5. Landing gear

$$W_{gear} = 0.045W_0 = 2,947 \text{ kg} \quad (40)$$

7.6. Surface controls

$$W_{controls} = 0.4W_0^{0.684} = 794 \text{ kg} \quad (41)$$

7.7. Total aircraft structure

$$W_{structure} = W_{wing} + W_{tail} + W_{fuselage} + W_{nacelle} + W_{gear} + W_{controls} = 28,187 \text{ kg} \quad (42)$$

7.8. Propulsion system

$$W_{propulsion} = 1.43W_{engine} = 6,226 \text{ kg} \quad (43)$$

7.9. Fixed equipment

$$W_{equipment} = 0.08W_0 = 5,299 \text{ kg} \quad (44)$$

7.10. Aircraft empty mass

$$W_{empty} = W_{structure} + W_{propulsion} + W_{equipment} = 39,712 \text{ kg} \quad (45)$$

7.11. Operational empty mass

$$W_{OEM} = W_{empty} + W_{crew} = 40,632 \text{ kg} \quad (46)$$

7.12. Aircraft zero-fuel mass

$$W_{ZFM} = W_{OEM} + W_{payload} = 50,232 \text{ kg} \quad (47)$$

7.13. Fuel mass

$$W_{fuel} = W_0 - W_{ZFM} = 16,001 \text{ kg} \quad (48)$$

7.14. Aircraft mass statement

After calculating the masses, the W_{fuel} (16,001 kg) was less than $W_{f_{required}}$ (21,764 kg). Therefore W_0 was increased and the calculations were re-done until the W_{fuel} and $W_{f_{required}}$ converged. This yielded a final W_0 and W_{fuel} of 87,280 kg and 25,149 kg respectively. A summary of the final masses can be found in Table 7.1 below.

Table 7.1: Final mass statement of initial design.

Component	Mass [kg]
Wing, W_{wing}	8,728
Tail, W_{tail}	1,746
Fuselage, $W_{fuselage}$	18,787
Nacelle, $W_{nacelle}$	1,945
Gear, W_{gear}	3,884
Surface Controls, $W_{controls}$	959
Structure, $W_{structure}$	36,048
Propulsion, $W_{propulsion}$	8,580
Fixed equipment, $W_{equipment}$	6,982
Empty Mass, W_{empty}	51,611
Operational items	400
Crew	520
Operational Empty Mass, W_{OEM}	52,531
Passengers	8,000
Baggage	1,600
Zero Fuel Mass, W_{ZFM}	62,131
Fuel, W_{fuel}	25,149
Max Take-Off Mass, W_0	87,280

8. Performance

8.1. Range

Rather than checking if the design could achieve the requirements, the requirements were fixed, and the aircraft features were designed to exactly meet these requirements. This can be demonstrated by re-calculating the range using our final parameters and the Breguet range Equation 49.

$$R = \frac{V}{C} \frac{L}{D} \ln \left(\frac{W_3}{W_4} \right) = 6000 \text{ nmi} \quad (49)$$

8.2. Cruise/climb

The thrust required to climb at the required rate of 300 *fpm* can be calculated using Equation 50 from [1].

$$T_{climb\ req} = W_{cruise} \left(\frac{V_v}{V_h} + \frac{1}{L/D_{cruise}} \right) = 46,056\ N \quad (50)$$

The thrust available at cruise altitude can be calculated using Equation 51 from [1].

$$T_{cruise} = 0.6987T^{0.9}e^{0.02(BPR)} = 70,909\ N \quad (51)$$

T_{cruise} is larger than $T_{climb\ req}$; therefore, satisfying the requirement.

8.3. Landing length

During landing it is common to deploy flaps and slats which change the shape of the wing to increase lift and drag. Although this decreases efficiency, it also lowers the stall speed of the aircraft allowing it to land more safely at a lower velocity and on shorter runways. The coefficient of lift during landing ($C_{L_{max}}$) was estimated using figures from [2] which yielded a value of 2.4 for double slotted flaps common on airliner wings.

To facilitate landing at 90% of airfields, a landing field length of 1400 *m* was chosen. The W/S required to enable this while meeting FAR 25 requirements was calculated using Equation 52 from [2].

$$W/S = \frac{S_{landing}\sigma C_{L_{max}}}{5} - S_a = 367\ kgm^{-2} \quad (52)$$

Where σ is the density ratio of the atmosphere at the airfield to that at sea level which assumed to be 1. S_a is the obstacle clearance distance which is 305 *m* for and airliner type aircraft.

This W/S was then used to iterate the calculations once again.

8.4. Take-off length

To facilitate take-off from 90% of airfields, a take-off field length of 1400 *m* was chosen. Based on figures from [2] the take-off parameter (TOP) for a dual engine aircraft for this field length was determined using figure 5.4 in [2]. The T/W required to enable this while meeting FAR 25 requirements was calculated using Equation 53 from [2].

$$T/W = \frac{W/S}{(TOP)\sigma C_{L_{TO}}} = 0.292 \quad (53)$$

Where the coefficient of lift at take-off ($C_{L_{TO}}$) is 80% of the $C_{L_{max}}$ as described in [2].

To enable this thrust to weight ratio with the updated W_0 calculated in section 7, more powerful engines are required. The required thrust was re-calculated to be 125,195 *N* per engine; therefore, the LEAP-1A with a T of 143,050 and a SFC_{cruise} of 0.51 */hr* is most suitable. This gives a final T/W of 0.334 for the aircraft.

9. Constraint analysis and optimisation

A sizing matrix plot was created to find a balance between thrust-to-weight and wing loading so that all crucial requirements were met, and these parameters were optimised. Three requirements were chosen to the subject of this parameter optimisation: landing field length, take-off field length, and cruise climb. The landing field length requirement is a subject of wing loading only, therefore appearing as a vertical line on the plot dictating a maximum wing loading constraint. The constraint is described in [2] and rearranged to Equation 54.

$$\frac{W}{S} = \frac{\sigma C_{L_{max}}}{0.85 \times 5} \left(\frac{S_{Land}}{1.67} - 305 \right) \quad (54)$$

The field length requirement was kept as 1400 *m* resulting in a maximum wing loading of 302.

The take-off field length is a function of variables $\frac{W}{S}$ and $\frac{T}{W}$ and has a linear relationship which is shown in Equation 55.

$$\frac{T}{W} = \left(\frac{W}{S} \right) \frac{1}{C_{L_{TO}} TOP} \quad (55)$$

Where $C_{L_{TO}}$ is the coefficient of lift at take-off and TOP is the take-off parameter.

To establish a relationship between thrust-to-weight and wing loading according to the cruise climb requirement of 300*fpm*, multiple intermediate steps were taken. First, cruise thrust was defined as a function of wing loading according to Equation 56.

$$T_{cruise} = 0.96 \left(\frac{W}{S} \right) S_{ref} \left(\frac{V_v}{V} + \frac{1}{D} \right) \quad (56)$$

As the lift to drag ratio is also dependent on wing loading, this relationship is not linear. Cruise thrust values were subsequently converted to thrust at sea-level by Equation 57.

$$T = \left[\left(\frac{T_{cruise}}{0.698} \right) e^{-0.02 \cdot BPR} \right]^{\frac{1}{0.9}} \quad (57)$$

Each thrust value was divided by its corresponding W_0 to find thrust-to-weight, which was calculated from multiplying wing loading by reference area. Figure 9.1 displays the constraint diagram with all three relationships, as well as the original design and optimised points.

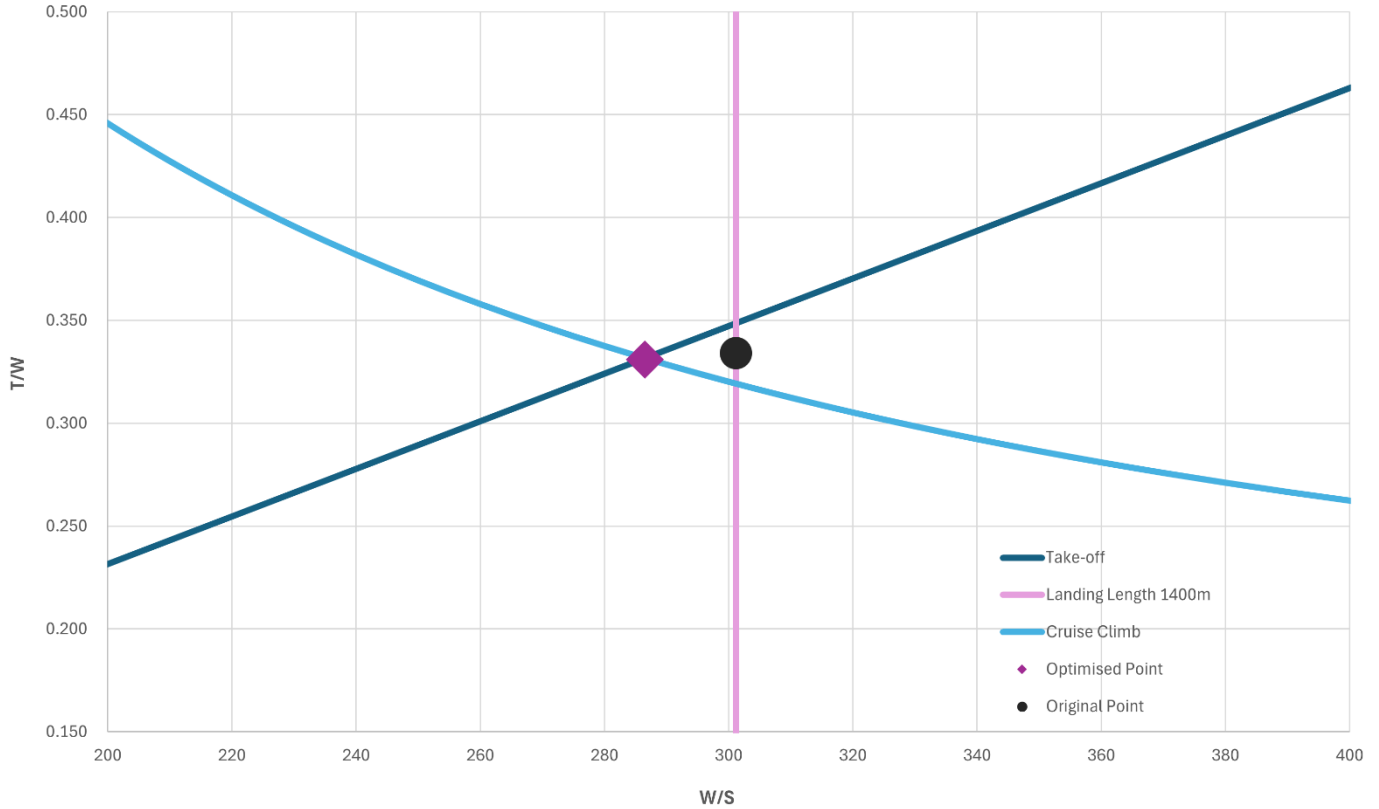


Figure 9.1: Constraint diagram.

By setting the optimised point at the lowest point of the constraint diagram, the new T/W and W/S were found to be 0.33 and 287 respectively. The lowest point on the constraint diagram reflects a balance between the specified performance parameters. Both T/W and W/S are minimised to use the least amount of thrust and retain optimal wing area without sacrificing on any critical flight requirements. Other performance metrics such as drag, fuel efficiency and structural weight are also optimised indirectly as a result of these improvements.

From the new T/W and W/S values, new aircraft geometry, masses and performance metrics were calculated using previous methods. As the original thrust capabilities were retained from keeping the same engines, the result of decreasing T/W was that the mass of the aircraft increased by 1%. This combined with a decrease in wing loading meant that the reference area increased significantly. This increased the dimensions of the wing, and subsequently the tail dimensions to maintain aerodynamic stability. The optimised geometry is presented in Table 9.1. In order to reduce the effect of increased wing area on the wingspan, the aspect ratio was reduced from 10 to 9. This reduced the wingspan from a possible 56m to 52.6m which makes it easier to park at smaller airports.

Table 9.1: Summary of optimised aircraft geometry.

Fuselage length [m]	40.6		
Passenger cabin length [m]	22.3		
Fuselage external diam. [m]	3.7		
	Wing	Horizontal tail	Vertical tail
Area, S [m ²]	307.5	88.3	65.3
Aspect ratio, A [-]	9	9	9
Taper ratio, λ [-]	0.3	0.3	0.3
Span, b [m]	52.6	19	11.3
Mean chord, $\bar{c}$ [m]	6.4	4.6	5.8
Leading edge sweep, Λ_{LE} [°]	30.0	35.0	40.0
¼ chord sweep, $\Lambda_{c/4}$ [°]	27.6	29.7	35.0

Table 9.2: Mass statement of optimised design.

Component	Mass [kg]
Wing, W_{wing}	8,811
Tail, W_{tail}	1,762
Fuselage, $W_{fuselage}$	18,787
Nacelle, $W_{nacelle}$	599
Gear, W_{gear}	3,921
Surf. Controls, $W_{controls}$	965
STRUCTURE, $W_{structure}$	34,845
Propulsion, $W_{propulsion}$	8580
Fixed equip., $W_{equipment}$	7,049
A/C EMPTY, W_{empty}	50,473
Operational items	400
Crew	520
OEM, W_{OEM}	51,393
Passengers	8,000
Baggage	1,600
ZFM, W_{ZFM}	60,993
Fuel, W_f	27,116
MTOM, W_0	88,109

Re-calculating the range using Equation 58, the Breguet Range Equation from [2] confirms that the fuel weight is more than required.

$$R = \frac{V}{C} \frac{L}{D} \ln \left(\frac{W_2}{W_3} \right) = \frac{291.6}{0.00014167} (21.33) \ln(0.7763) = 6003 \text{ nmi} \quad (58)$$

Additionally, the fuel tank capacity was re-calculated to be 62,362 kg, using the new aircraft geometry and the method outlined in section 5.2. This is also more than enough to hold the 27,116 kg of fuel required.

10. Optimised aircraft layout drawing

The optimised aircraft was 3D modelled in SOLIDWORKS and a drawing showing the key dimensions can be seen in Figure 10.1 below. A 3D file of the model can be found at https://drive.google.com/file/d/14N4_L7mz18hXsLKy8cu1qleq5y-aREW/view?usp=sharing.

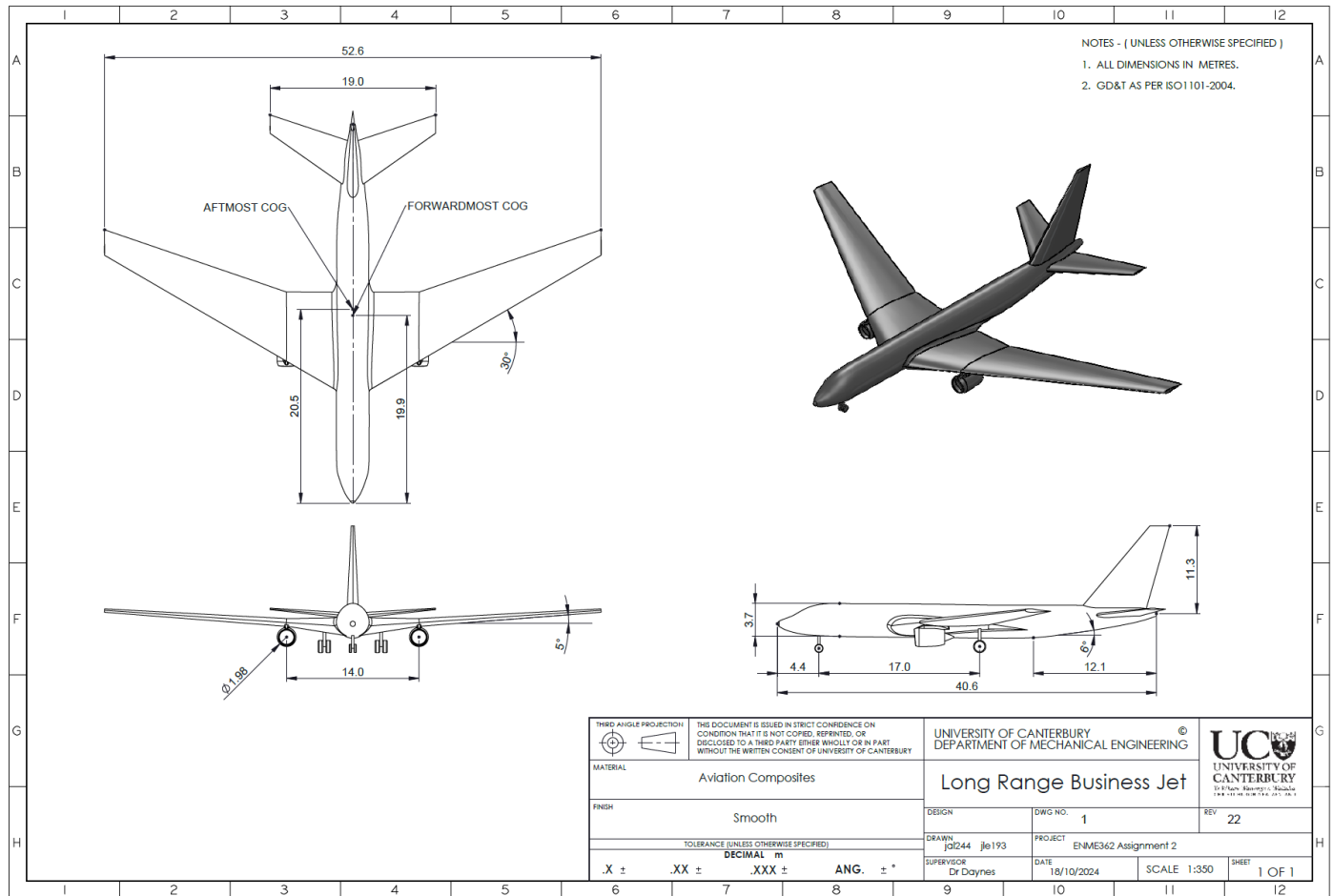


Figure 10.1: Optimised aircraft layout.

10.1. Wing position

The wing position was determined in its relationship to the centre of gravity (CoG). A group weights method was employed to find the most forward and aft CoGs for the aircraft. The centre of gravity along the aircraft's centreline (in the x direction) was calculated using Equation 59.

$$X_{cg} = \frac{\sum_{i=1}^n W_i X_i}{\sum_{i=1}^n W_i} \quad (59)$$

Where W_i is the combined weight of a weight group and X_i is its position in relation to the datum point located on the nose. The same method was used to find Y_{cg} but using the position of the components from the datum of the underside of the fuselage. No study was conducted for Z_{cg} as it was assumed the weight distributed would be near symmetric across the centreline.

The wing weight group consisted of the wing, nacelle, propulsion and fuel masses. The other weight groups were the payload, which included personnel and baggage, as well as tail weight and fuselage weight. The two weight groups which are variable during and between aircraft operation are payload and wing weight (due to fuel consumption). The combination which saw the calculation of the most forward CoG was the case of zero fuel and maximum payload, and the most aft CoG was determined to be the case of zero fuel and no payload (only two pilots).

According to [16], the CoG should fall on the centreline at the point corresponding to around 0.3 MAC (mean aerodynamic chord) of the wing. A convergence method was employed by varying the position of the wing along the x direction until the ideal CoG from [16] fell between the most forward and aft CoGs calculated using the weight group analysis. Table 10.1 and 10.2 shows the weights group analysis of the CoGs.

Table 10.1: Most Forward CoG Weight Group Analysis

Weight Group	W_i [kg]	X_i [m]	Y_i [m]	$W_i X_i$ [kgm]	$W_i Y_i$ [kgm]
Payload	10,520	15.0	1.1	157,800	11,551
Wing	18,955	20.0	0.7	378,529	13,875
Tail	1,762	35.0	3.7	61,676	6,450
Fuselage	53,631	20.3	1.8	1,087,715	98,145

Table 10.2: Most Aft CoG Weight Group Analysis

Weight Group	W_i [kg]	X_i [m]	Y_i [m]	$W_i X_i$ [kgm]	$W_i Y_i$ [kgm]
Pilots	200	2.0	1.4	400	280
Wing	18,955	20.0	0.7	378,529	13,875
Tail	1,762	35.0	3.7	61,676	6,450
Fuselage	53,631	20.3	1.8	1,087,715	98,145

The final CoGs are presented in Table 10.3.

Table 10.3: CoG Analysis Results

CoG Type	X [m]	Y [m]
Most Forward	19.9	1.5
Most Aft	20.5	1.6
0.3 MAC	20.0	—

The 0.3MAC CoG fell between the most forward and aft CoGs, but favoured the most forward, as this was the case with the max payload which is more likely to be the standard operational case. The analysis determined that the wing should be positioned so that the 0.3MAC point is placed 20m from the nose.

10.2. Tire sizing

Tires had to be selected for the landing gear wheels, thus a study of the minimum size requirements was conducted. First, the gear arrangement had to be determined so the number of wheels could be finalised. According to requirements from [1], a conventional tricycle arrangement with two wheels per strut was chosen as the max gross weight fell in the range of 50,000 to 200,000 lb (W_0 was 194,000 lb). The diameter and width of the wheels were calculated using Equations 60 and 61 respectively.

$$D = 5.3W_w^{0.315} \quad (60)$$

$$W = 0.39W_w^{0.48} \quad (61)$$

Where W_w is the weight on each wheel in kilograms. According to [1], the main tires carry around 90% of the aircraft's weight. The nose gear thus carries 10% of the aircraft's weight, as well as a further dynamic load caused by braking on landing. This braking load is a function of the aircraft's deceleration on landing, which is specified in [1] to be 3 [m/s^2], and the position of the landing gear which is discussed in the following section of this report. To increase the margin of safety, and allow for later aircraft growth, the weights were evaluated at the max gross weight and the loads were increased by 25%. Table 10.4 summarises the loads and calculated dimensions of the landing gear wheels.

Table 10.4: Calculated Wheel Weights and Dimensions

Wheel Group	W_w [kg] (+ 25% and DBL for Nose Gear)	Diameter [inches]	Width [inches]
Nose	7,141	34.2	10.9
Main	24,781	50.5	19.7

The calculated dimensions from Table 10.4 were then used as a guide to find suitable tires from a manufacturer. The Michelin aircraft tire catalogue [17] was used to find a suitable set of tires to fit the dimensions. Table 10.5 shows the tire selection for the main and nose gear wheels.

Table 10.5: Finalised Tire Selections from [17].

Wheel Group	Part No.	Size (DXW) [inches]
Nose	M10001	34X10
Main	M42203	50X20

The diameters of the Michelin tires selected fall just under the calculated specifications. However, as there is an additional load factor of 25% built into these calculations, this does not compromise on the safety of the gear.

10.3. Landing gear position

The landing gear must be positioned on the aircraft so that it meets certain loading and storage requirements. The front nose gear must take around 10% of the aircraft's gross weight, while the main gear takes the other 90%. This means that the nose gear must be placed around nine times further in front of aircraft's centre of gravity than the main gear is behind the CoG. These load requirements are specified in Equation 62 and 63 to account for the shifting CoG throughout and between flights. The calculated results of these equations for the final parameters are also displayed to show compliance with this standard.

$$M_a/B = 0.053 > 0.05 \quad (62)$$

$$M_f/B = 0.090 < 0.20 \quad (63)$$

Where M_a and M_f is the distance between the main wheels and aftmost and forwardmost CoG respectively, and B is the distance between nose and main gear. Table 10.6 details the finalised landing gear positions.

Table 10.6: Landing Gear Positions

Parameter	Distance between	Value [m]
B	Nose to main gear	17.00
N_a	Nose to aft CoG	16.10
N_f	Nose to forward CoG	15.46
M_a	Main to aft CoG	0.90
M_f	Main to forward CoG	1.54
H	CoG and ground	3.30

The height of the landing gear relates to the parameter H , and this was chosen to ensure that there was sufficient clearance between the ground and the bottom of the nacelles. From [1], sufficient A and B angles were chosen to place the nose gear in a position so that the nacelles are safe from debris thrown up from the wheels. The A angle is the top-down angle between the nose gear and nacelles, and B is the vertical angle. These were chosen to be 30° and 10° respectively. The spacing between the main gear is to enable them to be stored inside the undercarriage below the wings after folding inwards.

After finding the positioning of the gear, the dynamic load could be determined for the nose gear on landing. The dynamic load was calculated using Equation 64.

$$(Dynamic\ breaking\ load)_{nose} = \frac{3HW}{gB} = 5,230 [kg] \quad (64)$$

Where the constant 3 represents the deceleration specification, and g is acceleration due to gravity. The total breaking load was evenly divided across two nose wheels to 2,615 kg each. This load was added to the W_w value for the nose gear in Table 10.4 to calculate the necessary tire dimensions.

11. Flight envelope

11.1 Basic flight envelope

At altitude the dynamic pressure-based airspeed needed for load calculations is less than the actual airspeed at altitude due to the decrease in air density. Therefore, an equivalent velocity is used for load estimation and can be calculated using Equation 65 below.

$$V_{equivalent} = \sqrt{\frac{\rho}{\rho_{sl}}} V_{actual} \quad (65)$$

Where ρ is the air density at altitude of 0.3796 kgm³ and ρ_{sl} is the air density at sea level of 1.225 kgm³ and V_{actual} is the actual air speed calculated using Equations 66 and 67 depending on whether the load factor is positive or negative.

$$V_{actual} = \sqrt{\frac{Wgn}{\rho SC_{Lmax+ve}}}, n \geq 0 \quad (66)$$

$$V_{actual} = \sqrt{\frac{Wgn}{\rho SC_{Lmax-ve}}}, n \leq 0 \quad (67)$$

Where $C_{Lmax+ve}$ and $C_{Lmax-ve}$ are +1.91 and -1 respectively, chosen based on another similar aircraft in [1]. The positive G and negative G stall speeds (V_{s+1g} and V_{s-1g}) were calculated with at a load factor of 1 and -1 respectively.

The dive velocity was calculated as $1.4V_{cruise}$ as is commonly done in [1].

The V-n diagram for the manoeuvre flight envelope at altitude was constructed by plotting the equivalent velocity over a range of load factors from -1.3 to $+2.6$ as is required by regulatory bodies. The negative load factor was calculated as -0.5 times the positive load factor which was chosen based on a similar aircraft in [1]. Connecting these curves to the equivalent dive velocity with two straight parallel lines encloses the manoeuvre flight envelope that can be seen in Figure 11.1.

11.2 Gust loads flight envelope

Gusts tend to follow a cosine-like intensity increase as the aircraft flies through it; therefore, a statistical gust alleviation factor (K) is used instead and can be calculated using Equation 68 below.

$$K = \frac{0.88\mu}{5.3 + \mu} = 0.6495 \quad (68)$$

Where μ is the mass ratio and can be calculated using Equation 69.

$$\mu = \frac{2(W/S)}{\rho g \bar{c} C_{L\alpha}} = 14.93 \quad (69)$$

Where $C_{L\alpha}$ is the lift curve slope and can be calculated using Equation 70.

$$C_{L\alpha} = \frac{2\pi A}{2 + \sqrt{4 + \frac{A^2 \beta^2}{\eta^2} \left(1 + \frac{\tan^2(\Lambda_{LE})}{\rho^2}\right)}} = 5.58 \text{ [rad}^{-1}\text{]} \quad (70)$$

Where η is an efficiency of 95% as described by [16], and β is described by Equation 71 below.

$$\beta = 1 - M^2 \quad (71)$$

Using K and $C_{L\alpha}$ the load factor can be calculated for various vertical gust velocities (U_{de}) using Equation 72 below.

$$n = 1 \pm \frac{KU_{de}V\rho SC_{L\alpha}}{2W} \quad (72)$$

The U_{de} at dive velocity and at cruise velocity were determined to be 5 ms^{-1} and 12 ms^{-1} respectively using a figure from [1]. Calculating n using these values and connecting them with strait lines on the V-n diagram yields the completed gust loads flight envelope seen on Figure 11.1.

11.3 Combined flight envelope

The gust loads flight envelope is fully contained within the manoeuvre flight envelope; therefore, the combined flight envelope is the same as the manoeuvre flight envelope as seen on Figure 11.1 below.

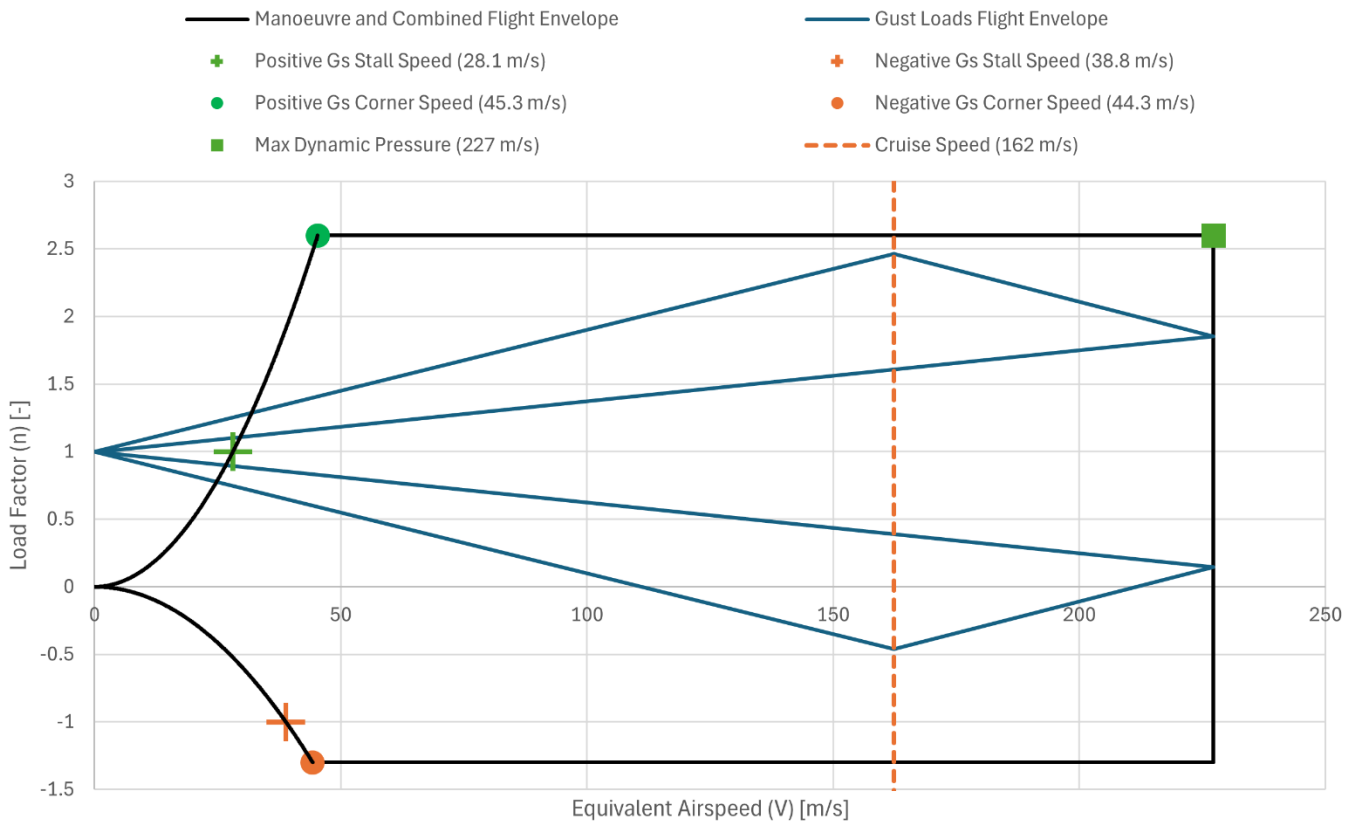


Figure 11.1: V-n diagram of basic, gust loads and combined flight envelopes.

The V-n diagram shows the limit load factor as a function of the equivalent airspeed (EAS) at cruise altitude. The positive and negative G stall speeds are the slowest EAS's that the aircraft can fly while upright and inverted respectively. Similarly, the corner speeds are the slowest EAS's at which the aircraft can reach the maximum load factors without stalling i.e. the max angle of attack (MAXAOA) while upright and inverted. The solid vertical line on the right of the diagram represents the dive speed of the aircraft and is where the maximum dynamic pressure is experienced (max q).

The gust loads flight envelope is completely encased within the manoeuvre flight envelope which is typical for larger aircraft such as this. Smaller, lighter aircraft are often more affected by gust loads and therefore the combined flight envelope would need to be altered to account for this; however, our aircraft does not require it.

12. Trade-off study

At the start of this assignment, the cruise velocity and max take-off weight were rather arbitrarily chosen as Mach 0.85 and approximately 65,000 kg based on current similar aircraft. This trade-off study explores how changes in cruise velocity might change the max take-off weight, and how iterations of this weight calculation cause it to converge to a more accurate value.

Using the statistical data for max take-off weight (W_0) and empty weight fraction (W_e/W_0) from Table 4.1, a figure was created from which the line of best fit has the form of Equation 73.

$$W_e/W_0 = AW_0^C = 0.512 \quad (73)$$

Where C is -0.02 , A is 0.64 and W_0 is an initial guess of 65,000 kg. To calculate the fuel fraction ($W_{f_{required}}/W_0$), mission segment weight fractions (MSWFs) were required. The warmup and take-off (W_1/W_0), climb (W_2/W_1), and landing (W_5/W_4) MSWFs were historically determined in [2] as 0.970, 0.985, and 0.995 respectively. The cruise MSWF was calculated using Equation 74, the Breguet Range Equation from [2].

$$W_3/W_2 = \exp \left[\frac{-RC}{V(L/D_{cruise})} \right] \quad (74)$$

Where R is the range in nautical miles, C is specific fuel consumption (SFC), V is cruise velocity and L/D_{cruise} is the lift-to-drag ratio at cruise. The SFC of a LEAP1-A is $0.51/hr$ and the L/D was calculated using the component build up method described in section 6.1 of this report.

The loiter MSWF was calculated using the endurance Equation 75 from [2].

$$W_4/W_3 = \exp \left[\frac{-EC}{L/D} \right] \quad (75)$$

Where E is the endurance/loiter time which is an FAA requirement found to be 45 minutes from [13].

Multiplying all the MSWFs together gives an overall MSWF (W_5/W_0) which could be used to calculate the W_f/W_0 using Equation 76 from [2] which includes a typical 6% allowance for reserved and trapped fuel.

$$W_{f_{required}}/W_0 = 1.06(1 - W_5/W_0) = 0.335 \quad (76)$$

These results could then be used to calculate the max take-off weight using Equation 77 from [2].

$$W_0 = \frac{W_{crew} + W_{payload}}{1 - W_{f_{required}}/W_0 - W_e/W_0} \quad (77)$$

The process was iterated until the W_0 calculated using Equation 73 and Equation 77 converged.

Once the weight for a specific cruise velocity was calculated it was used to re-do the optimisation study as described in section 9 before being plotted. This process was done 20 times for cruise velocities from Mach 0.7 to Mach 0.9 to observe the smooth polynomial relationship seen on Figure 12.1 below.

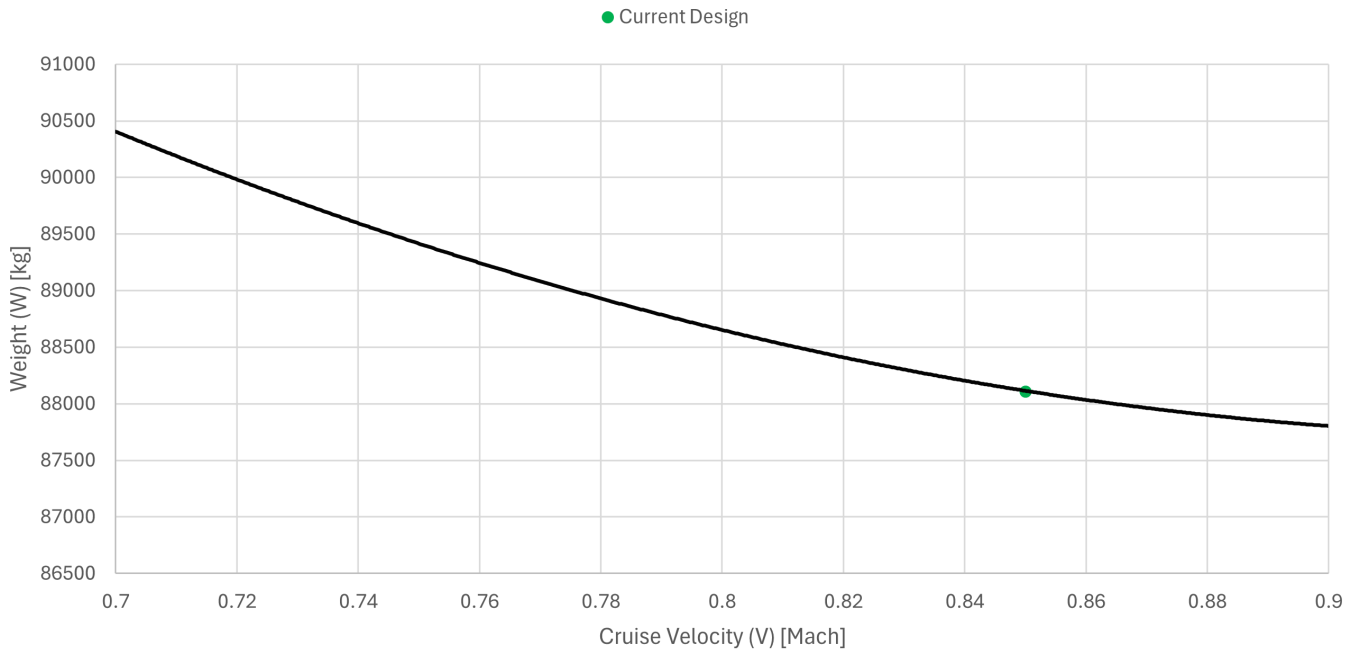


Figure 12.1: Trade-off study for cruise velocity against weight.

A clear trend emerged showing that the weight decreases with increased cruise velocity, but the rate of this decrease slows down towards higher cruise velocities. Increasing the cruise velocity could offer a slight improvement to our design; however, it is important to note that as the cruise velocity approaches Mach 1, the chance of experiencing supersonic airflow due to wind gusts increases. This is why the trade-off study only includes cruise velocities up to 0.9 Mach, as this was decided to be the upper limit before encountering supersonic flow too regularly.

13. Conclusions and further work

Using the T/W, W/S carpet plot, the design was optimised to meet all three specified requirements including cruise climb, and landing and take-off lengths. Figure 9.1 shows that the point representing the original design was below the take-off field length requirement, and exceeded the cruise climb specification. The means that the original aircraft was overpowered in cruise, but did not meet the lift requirements at lower speeds to safely take off from the desired runway lengths.

To remedy this, the wing loading, and thrust-to-weight were decreased until both take-off field length and cruise climb requirements were met. The optimised T/W ensures that the aircraft has sufficient thrust to achieve the required cruise climb rate without using excessive engine power, improving fuel efficiency during long-range flights. A result of the optimisation was that the weight of the aircraft increased from 87,280kg to 88,109kg. This increased weight was to allow for greater wing area

necessary to meet slow speed lift requirements for take-off performance. Larger wing area also increases the operational safety of the aircraft during take-off and landing as stall speed is reduced. Increased W_0 meant that the fuel weight capacity went from 24,149kg to 27,116kg to keep the range capacity at 6000nm which was a hard requirement for the project.

Using the original aspect ratio of 10, the wings would have been relatively long when compared to the aircraft's fuselage length. To reduce the wingspan, the aspect ratio was decreased to 9 for the wings and tail sections. This made the aircraft more proportional and more accessible for parking at smaller airports.

Overall, while the aircraft is heavier and larger than the original design, this was a necessary change to meet requirements which were conflicting in the previous iteration of the jet. The optimised design can now safely take off and land on runways meeting the 1400m specified length. This makes the jet suitable for operating on 90% of the world's runways, which was one of the main selling points of the aircraft.

There are several areas of work which could be explored further to increase the potential and accessibility of the design. The wingspan of the aircraft is still relatively large when compared to traditional business jets operating out of target airports. Not all airports which fall under the target field length specifications would have the gate infrastructure for an aircraft with a 50-metre wingspan. While the aircraft is parked at these airports which may lack suitable infrastructure, it would have to be boarded and disembarked with mobile stairs rather than a jet bridge at a gate, which could be inconvenient, especially in adverse weather conditions. If parking space limitations due to excessive wingspan persists to be a significant issue, work could be done to implement a folding wingtip design similar to current technology from Liebherr Aerospace exhibited in the recent Boeing 777X as detailed in [18]. However, a folding mechanism would increase the weight of the wing and add greater complexity to the design.

When more detailed work is done on the internal cabin of the aircraft, a modular cabin design should look to be implemented. As discussed previously, the ability to diversify the role of the aircraft is important to its market appeal. By designing the cabin to be interchangeable between business class seating, VIP charter, or government/military transport, the number of potential customers for this unique jet would increase significantly. A modular cabin would also mean that the manufacturing of the aircraft remains unchanged excluding the installation of each particular cabin type.

Business class passengers desire more comfort out of their flying experience, which is why further work would involve research and implementation of noise reduction technologies. Most jet aircraft noise comes from their engines, as turbulent mixing of exhaust gases with the atmosphere creates shearing which produces the high-pitched piercing noise we are familiar with. To reduce high frequency noise, the exhaust pattern can be modified by changing the geometry of the nozzle. According to [19], adding chevrons (serrated edges) to the aft of the nacelle can enhance the mixing of exhaust gases with the atmosphere, and reduce high speed shearing resulting in a quieter aircraft. A significant amount of the noise also comes from the rotation of the fan and compressor stages within the engine. Special acoustic panels and liners can be placed on the inside of the fan casing to dampen the vibrations caused by the rotating elements, providing effective noise suppression.

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